Lesson 3 — Operators & Input/Output

Topic

Operators & Input/Output

Definition

Operators are special symbols that perform operations on variables and values. Python provides Arithmetic, Comparison, Logical, and Assignment operators to manipulate data.

Input/Output functions allow programs to communicate with users. The `input()` function reads data from the keyboard, and the `print()` function displays output to the screen.

Detailed Meaning

Arithmetic operators perform math: `+`, `-`, `*`, `/` (float), `//` (floor), `%` (remainder), `**` (power).

Comparison operators return boolean: `==`, `!=`, `>`, `<`, `>=`, `<=`.

Logical operators combine conditions: `and` (both True), `or` (one True), `not` (inverts).

Assignment operators: `=` assigns. `+=`, `-=`, `*=`, `/=`, `//=`, `%=`, `**=` do operation and assign together.

`input()` returns a string. Use `int()` or `float()` to convert. `print()` displays output — use `sep` for separator and `end` for ending (default newline).

Example

```python
# Arithmetic
a, b = 15, 4
print(a + b, a - b, a * b, a / b, a // b, a % b, a ** b)

# Comparison
print(10 == 10, 10 != 5, 10 > 5, 10 < 5)

# Logical
print(True and False, True or False, not True)

# Input/Output
name = input("Enter name: ")
age = int(input("Enter age: "))
print(f"Hello {name}, you are {age} years old")
```

Code Example

```python
# Operators & Input/Output

# ========== ARITHMETIC ==========
print("--- Arithmetic Operators ---")
```

```
x, y = 20, 6
print(f"{x} + {y} = {x + y}")
print(f"{x} - {y} = {x - y}")
print(f"{x} * {y} = {x * y}")
print(f"{x} / {y} = {x / y:.2f}")
print(f"{x} // {y} = {x // y}")
print(f"{x} % {y} = {x % y}")
print(f"{x} ** {y} = {x ** y}")

# ========== COMPARISON ==========
print("\n--- Comparison Operators ---")
a, b = 10, 20
print(f"{a} == {b}: {a == b}")
print(f"{a} != {b}: {a != b}")
print(f"{a} > {b}: {a > b}")
print(f"{a} < {b}: {a < b}")
print(f"{a} >= {b}: {a >= b}")
print(f"10 < 20 < 30: {10 < 20 < 30}")

# ========== LOGICAL ==========
print("\n--- Logical Operators ---")
p, q = True, False
print(f"True and False: {p and q}")
print(f"True or False: {p or q}")
print(f"not True: {not p}")
```

```python
age = 25
has_id = True
if age >= 18 and has_id:
    print("Entry allowed")

# ========== ASSIGNMENT ==========
print("\n--- Assignment Operators ---")
n = 50
print(f"n = {n}")
n += 10; print(f"n += 10 -> {n}")
n -= 5; print(f"n -= 5 -> {n}")
n *= 2; print(f"n *= 2 -> {n}")
n //= 3; print(f"n //= 3 -> {n}")
n %= 7; print(f"n %= 7 -> {n}")
n **= 2; print(f"n **= 2 -> {n}")

# ========== INPUT/OUTPUT ==========
print("\n--- Input/Output ---")
name = input("Enter your name: ")
age = int(input("Enter your age: "))
print(f"\nName: {name}, Age: {age}")

print("Hello", "World", sep="-")
print("No newline", end=" ")
print("on same line")
```

```
# ========== CALCULATOR ==========
print("\n--- Calculator ---")
num1 = float(input("Enter first number: "))
op = input("Enter operator (+, -, *, /): ")
num2 = float(input("Enter second number: "))

if op == "+":
    result = num1 + num2
elif op == "-":
    result = num1 - num2
elif op == "*":
    result = num1 * num2
elif op == "/":
    result = num1 / num2 if num2 != 0 else "Error: Division by zero"
else:
    result = "Invalid operator"

print(f"Result: {result}")
```

Code Explanation

Arithmetic operators: Floor division `//` rounds down to integer. Modulus `%` gives remainder. Power `**` raises to exponent:

```python
print(17 // 5)  # 3
print(17 % 5)   # 2
print(2 ** 10)  # 1024
```

Comparison operators return `True` or `False`. Chained comparison `10 < 20 < 30` works as `10 < 20 and 20 < 30`.

Logical operators combine boolean expressions. `and` needs both True, `or` needs one True, `not` inverts the value.

Assignment operators are shorthand. `n += 10` is same as `n = n + 10`. All compound operators work similarly.

`input()` returns a string. Wrap with `int()` or `float()` for numbers. **`print()`** displays output. Use `sep` to change separator (default space) and `end` to change ending (default `\n`):

```python
print("A", "B", sep="-", end="!")  # A-B!
```